

Ambuj Tiwari

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PROFESSIONAL SUMMARY

Software Engineer with **4+ years of experience** in backend development, microservices, and cloud-native applications using **Java, Spring Boot, and REST APIs**. Currently working as a **System Development Engineer at Amazon**, supporting service reliability, JDK migration, and operational improvements. Experienced in designing and integrating enterprise APIs, implementing reactive and concurrent service interactions, and building serverless solutions on AWS. Hands-on experience with **Generative AI, LLMs, and RAG pipelines** for intelligent developer automation. Strong background in debugging, performance optimization, and delivering enterprise solutions in Agile environments.

TECHNICAL SKILLS

Languages: Java 8/11/17/21, Python, C, TypeScript, JavaScript, SQL

Backend Frameworks: Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Spring WebFlux, Hibernate

Microservices & APIs: Microservices Architecture, REST APIs, SOAP/WSDL, OpenAPI/Swagger, Design-First, WebClient

Frontend: Angular 12+, TypeScript, RxJS, Angular Material, HTML5, CSS3/SCSS, JavaScript

Databases: MySQL, DynamoDB, FAISS

Cloud / AWS: AWS Lambda, API Gateway, EC2, S3, SQS, Step Functions, Bedrock

DevOps & Tools: Docker, CI/CD, Jenkins, Maven, Git, Postman, SonarQube, Dynatrace, IntelliJ IDEA, VS Code, Linux

Testing: JUnit, Mockito, TDD

AI / GenAI: LLMs, RAG, LangChain, Anthropic Claude, Tree-sitter

Concepts: CompletableFuture, Reactive Programming, SOLID, Design Patterns

Methodologies: Agile, Scrum

PROFESSIONAL EXPERIENCE

Amazon — System Development Engineer

Bengaluru, India | Aug 2026 – Present

- Handle production support tickets and investigate service issues using logs, monitoring, and existing operational workflows.
- Contribute to **MCM traffic migration activities**, supporting the movement of service traffic from one region to another region.
- Support services during peak business events, including **Prime Day and Big Day Deals**, by monitoring service health and assisting with availability and operational issues.

Cognizant – Kolkata / Remote Mar 2022 – July 2026

Associate – Backend Developer | Aug 2024 – Present

- Enhanced and maintained **Spring Boot microservices**, reducing production issues by **15%**.
- Designed and implemented APIs using a **Design-First** approach and integrated **SOAP-based (WSDL)** services for seamless communication with enterprise systems.
- Engineered optimized service interactions using **CompletableFuture** for parallel processing and **Spring WebFlux WebClient** with **OpenAPI-generated clients**, ensuring reactive communication, type safety, and reduced manual integration effort.
- Built **AWS serverless workflows** (Lambda, API Gateway, DynamoDB), reducing manual effort by **25%**.
- Performed performance tuning and debugging to improve system reliability across services.
- Collaborated with cross-functional teams to deliver features across Agile sprints.

Programmer Analyst – Full Stack Developer | Mar 2024 – Nov 2024

- Engineered backend services using **Java** and **Spring Boot** for insurance workflows.
- Integrated **Angular** modules with backend **REST APIs** across **5+** screens.
- Optimized **SQL** queries, improving response time by **20%**.
- Participated in Agile ceremonies, ensuring smooth and predictable delivery.

Programmer Analyst Trainee – Frontend Developer | Aug 2022 – Mar 2024

- Built responsive UI components using **Angular** and **TypeScript**.
- Improved UI/UX consistency, reducing defects by **20%**.
- Collaborated with senior developers to deliver features efficiently.

Full Stack Engineering Intern | Mar 2022 – Aug 2022

- Completed training modules using **Java**, **Spring Boot**, **Maven**, and **SQL**.

KEY PROJECTS

Nexus-P2C-Migration

- Migrated and modernized existing services using **Spring Boot microservices** architecture and standardized API design practices.
- Implemented reactive service-to-service communication using **Spring WebClient** and leveraged **OpenAPI-generated clients** to reduce manual integration code.
- Designed concurrent workflows using **CompletableFuture** for optimized execution and reduced processing time.

- Integrated monitoring and distributed tracing via **Dynatrace**; supported **CI/CD** deployments with **SonarQube** quality and code-coverage checks.

KT-Buddy – AI-Powered Knowledge Transfer Assistant

- Developed an AI assistant that transforms large codebases into interactive, explainable knowledge systems, reducing onboarding effort.
- Integrated **Anthropic Claude** to generate Javadoc summaries, inline code explanations, and contextual insights; built **RAG**-powered natural-language Q&A grounded in source code.
- Built repository ingestion with GitHub URL validation and automated cloning; used **Tree-sitter** for structured source-code parsing.
- Generated architecture summaries and flow visualizations with **ReportLab**; deployed end-to-end using **Python** and **Streamlit**.

GenAI Visionaries

- Built a serverless pipeline with **18+ AWS Lambdas**, integrating AWS services to reduce manual effort by **60%**.
- Applied **FAISS**-based semantic search with **LLMs** to improve developer productivity through automation.

Engineering Excellence

- Led migration of microservices to **Java 21**, improving runtime performance.
- Created lightweight dependencies to reduce external reliance and executed testing achieving **90%+** coverage.

G-Synergy

- Enhanced **Spring Boot microservices** and engineered **REST APIs** for insurance-workflow integrations.
- Optimized database performance by **20%**.

MNX Rationalization

- Built responsive UI components using **Angular** and **TypeScript** across **5+** screens, improving user experience and consistency.
- Resolved cross-browser compatibility issues to improve overall UI/UX.

EDUCATION

B.Tech in Electronics and Communication Engineering – CGPA: 8.7

RCC Institute of Information Technology | 2018 – 2022

CERTIFICATIONS

- Claude Certified Architect Foundations
- Spring Professional Certification
- MySQL and JDBC
- LangChain for Beginners

ACHIEVEMENTS

- Trained **100+** Gen-C engineers on **Java** and **Spring Boot**, earning formal recognition from management.
- Awarded manager Cheers Points for consistently completing and delivering multiple user stories ahead of expectations.